Paper II: best verified results available so far

Provisional evidence snapshot

NOT MANUSCRIPT-PROMOTED. This is a durable lower-bound snapshot of completed, exact-scored evidence. Live lanes remain unfinished; no missing result is interpreted as a failure or a zero.

What is already secured

Four full-horizon Hubbard–Holstein matched-accuracy pairs reach $t = 20$. Across those pairs, mean infidelity is 6.37×10^{-6}–1.12×10^{-5} for APM/exchange and 6.00×10^{-7}–2.08×10^{-6} for AVQDS. A separate completed AVQDS arm reaches 1.174×10^{-7}, but has no qualified APM mate.

Best individual $t = 20$ mean infidelity	1.174×10^{-7}, AVQDS, strong/fast parameter motion; **unpaired**
Best within a qualified matched pair	6.004×10^{-7}, AVQDS, fast/weak parameter-motion pair
Best observed $t = 20$ APM mean infidelity	6.370×10^{-6}, weak/slow parameter-motion pair
Matched-pair coverage	4 qualified pairs / 8 method arms, all reproduced on page 2
Observed logical-cost pattern	Within these four retained pairs, APM's recorded graph tuple and N_θ are componentwise smaller than AVQDS's

Interpretation boundary. These are observed, exact-referenced trajectories under the recorded response-normalized matched-accuracy contract. They establish an accuracy floor and a reproducible resource trend for the completed pairs; they do not establish universal method superiority or completion of the 39-cell prospective matrix.

Evidence semantics

- Accuracy quantities are physical errors against exact references; exact evolution is reporting-only.
- The resource tuple is $(\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q}, N_\theta)$, a routing-independent logical/graph proxy.
- Qiskit compilation is excluded from this prospective campaign; implementation-parity evidence is therefore **OUT OF SCOPE**, not missing success evidence.

First-page evidence manifest

Evidence block	Rows	Audit / state	Scientific status
Canonical HH staged ledger	345	**PASS**	4 qualified $t = 20$ matched pairs; 15 qualified and 21 pending method arms overall
Prospective HH	1 scored	**BOUND**	AVQDS $t = 5$ individually qualified; comparison/cost claim pending APM mate
SIAM $N_b = 4$ pilot	2 scored	**BOUND**	numerical candidates only; no individually qualified matched pair yet
Live prospective lanes	3 states	**OPEN**	Hubbard RUNNING; HH APM PREFLIGHT_ONLY; SIAM r07 SUBMITTED

Canonical HH cluster set (31):

9697391, 9698201, 9698247, 9698264, 9701941, 9703334, 9703747, 9703851, 9705193, 9707629, 9708157, 9708462, 9708981, 9709593, 9709600, 9709707, 9709708, 9709713, 9709723, 9709731, 9709741, 9709744, 9709745, 9709750, 9709763, 9709772, 9709886, 9709947, 9709982, 9712867, 9713112.

Current prospective identities:

HH completed: `9715119.0`. Hubbard running at the last combined scheduler capture: `9714499.0`, `9714501.0`. SIAM scored: `9714484.0`, `9715147.0`; r07 submitted exactly once as `9715270` after that capture.

Durable sources:

Canonical report PDF SHA-256: `e0fdcef1aec522d4d3bfd54e9563842e8847f4608975a389946a60b010644929.`
Canonical report JSON SHA-256: `6461dc8255fb68276932bd2c7856aa43d856c0300d3f199b2e5d95305f5c8da9.`
Latest combined capture SHA-256: `64deab5ede574db7f29188913ea2fb7ed938669d068351cd06a96f3875a2476e.`

Full file hashes, source paths, claim bindings, and caveats are retained in the adjacent source-map and manifest JSON files. The canonical 14-page campaign report and one-page SIAM r06 report are appended unchanged after this three-page cover.

Completed full-horizon HH evidence: four matched $t = 20$ pairs

The table reproduces the eight method arms identified by the canonical report as four qualified response-normalized matched-accuracy pairs. Errors are time-mean absolute errors against the exact reference. $\mathcal{B}$ is the recorded aggregate normalized accuracy burden; it is an accuracy quantity, not a hardware cost.

Regime	Pair definition	Method	Mean infidelity	Mean $\lvert\Delta E\rvert$	Mean $\lvert\Delta D\rvert$	Mean $\lvert\Delta C\rvert$	$\mathcal{B}$	Logical tuple
weak / slow	parameter motion	AVQDS	2.084×10^{-6}	1.471×10^{-4}	1.362×10^{-4}	7.953×10^{-4}	1.421	$(5214, 5476, 9281, 945)$
weak / slow	parameter motion	APM/exch.	6.370×10^{-6}	8.593×10^{-5}	2.157×10^{-4}	4.281×10^{-4}	1.233	$(172, 254, 365, 53)$
weak / slow	state + parameter	AVQDS	1.200×10^{-6}	1.040×10^{-4}	9.707×10^{-5}	5.793×10^{-4}	1.009	$(5130, 5390, 9048, 926)$
weak / slow	state + parameter	APM/exch.	6.395×10^{-6}	1.040×10^{-4}	1.903×10^{-4}	3.892×10^{-4}	1.232	$(178, 248, 395, 53)$
weak / slow	state motion	AVQDS	1.603×10^{-6}	1.506×10^{-4}	1.447×10^{-4}	7.919×10^{-4}	1.461	$(1908, 2074, 3642, 370)$
weak / slow	state motion	APM/exch.	7.360×10^{-6}	1.328×10^{-4}	1.228×10^{-4}	8.193×10^{-4}	1.303	$(182, 256, 388, 54)$
fast / weak	parameter motion	AVQDS	6.004×10^{-7}	4.242×10^{-4}	1.182×10^{-4}	3.856×10^{-4}	0.062	$(170, 238, 371, 61)$
fast / weak	parameter motion	APM/exch.	1.121×10^{-5}	5.605×10^{-4}	1.914×10^{-4}	4.872×10^{-4}	0.107	$(96, 144, 246, 40)$

Accuracy floor

- The completed matched-pair evidence is already in the 10^{-6}–10^{-5} mean-infidelity range; the best completed individual arm reaches 1.174×10^{-7}.
- The corresponding mean absolute observable errors shown here are in the 10^{-5}–10^{-4} range for energy/doublon and the 10^{-4} range for the displayed charge-imbalance response.
- The best mean infidelity is not used as a stand-alone winner: matched response burden and the retained observable caps remain part of qualification.

Resource observation

- For every completed pair above, APM/exchange records fewer logical one- and two-qubit graph operations, lower logical depth proxy, and fewer terminal independent parameters.
- This is a routing-independent proxy comparison. No Qiskit compilation, backend routing, shot cost, or hardware-noise claim is made here.
- Final manuscript attribution remains contingent on the prospective fixed-code matched-pair campaign and its pre-submit audit.

Best individual accuracy record (not a cost comparison): strong/fast AVQDS parameter-motion, cluster `9698264`, mean infidelity 1.1744525×10^{-7}, $N_\theta = 62$, graph tuple $(182, 256, 386)$. Its run SHA-256 is `e00618453d30fad48854eae7a462ab229bfa528f808efea97dbbbed7056ca540`. It is deliberately excluded from the four matched rows above because no qualified APM mate exists.

Source: canonical 345-record Paper-II staged campaign report. The complete report, including every evaluated row, safeguards, pending arms, definitions, and plots, begins on page 4 of this bundle.

Prospective evidence and the live boundary

New exact-scored trajectories

Benchmark	Method	Horizon	Cluster	Mean infidelity	$\mathcal{B}$	Status
HH $N_{ph} = 1$, crossover/midres	AVQDS, cut 10^{-2}	$t = 5$	9715119	1.658×10^{-5}	0.1119	**QUALIFIED**; APM mate pending, so no cost comparison
SIAM $N_b = 4$	AVQDS, r02 cut 10^{-3}	$t = 1$	9714484	2.602×10^{-5}	0.4380	CANDIDATE; broad guard passes, individual target fails
SIAM $N_b = 4$	APM/exchange, r06 cut 10^{-5}	$t = 1$	9715147	2.237×10^{-6}	1.5954	REJECTED PAIR; broad response guard fails

Verified detail

HH $t = 5$: mean $|\Delta E| = 3.315 \times 10^{-3}$, mean doublon error 9.671×10^{-4}, mean maximum site-occupation error 1.894×10^{-3}, and logical tuple $(270, 486, 617, 221)$. Exact source, seed, and source epoch are digest-bound; Qiskit compilation was not performed.

SIAM gate detail: AVQDS's worst primitive response-normalized mean error is 0.07938; APM r06's is 0.31897 in the small-response C_3 channel. Their mean $|\Delta E|$ values are 5.145×10^{-4} and 4.036×10^{-4}. The pair is accuracy-confounded and rejected; r07 is the pure planner's next APM rung.

Security statement. If every open lane stopped today, this bundle would still retain four complete HH $t = 20$ matched pairs, the unpaired 1.174×10^{-7} HH accuracy record, and one independently qualified prospective HH $t = 5$ trajectory, all with exact-reference accuracy and source hashes.

What this PDF does not claim

- It is not the final Paper-II table or 39-cell manuscript result set.
- It declares no SIAM winner and infers no threshold monotonicity.
- It reports no Qiskit-compiled, routed, shot, or hardware cost.
- It does not reopen the superseded adaptive FS-radius experiment.

Open work is explicit

Lane	Typed state	Consequence
HH APM mate	**PREFLIGHT**	dry-run only; no result
Hubbard L4	**RUNNING**	9714499/9714501; no score yet
SIAM r07	**SUBMITTED**	9715270; result pending
Full matrix	**INCOMPLETE**	final promotion not claimed

Audit anchors

- canonical campaign: **PASS**, 345 immutable rows;
- prospective HH: exact/seed/source identities joined and score digest verified;
- SIAM: exact source epoch and 101-point grid digest bound;
- latest combined scheduler capture: 2026-08-30 03:44:09 UTC;
- source evidence revision: `d5590449`.

Reader map. Pages 4–17 are the unchanged 14-page canonical campaign report. Page 18 is the unchanged SIAM r06 results sheet. The adjacent JSON sidecars bind every source and every headline claim to a SHA-256 digest.

Paper II staged campaign and matched-accuracy results

Campaign snapshot: 2026-08-28 13:37 CDT **CHTC (last verified):** 3 legacy jobs held; all 6 cluster-9713112 processes are terminal
Verified frontier: 345 ledger records = 157 t5 rungs + 188 t20 records
Incremental update: 2 resource-rescue screens qualified and 3 terminal unsupported-step outcomes were retained as safeguards
Canonical accuracy policy: 5% exact-response target, 15% comparison guard, and factor-two aggregate-and-primitive APM/AVQDS accuracy matching
Policy replay: 280 of 345 records qualify, 48 require a tighter activation cut, and 17 are safeguards
t20 status: 138 evaluated continuations qualify, 48 require extension, and 2 threshold-floor safeguards are recorded; the matrix has 15 qualified and 21 pending slots
Active frontier: 9703334.0, 9703747.0, and 9703851.0 are held; 9713112.3 is transfer-pending after exit 3
Access status: process 3 is remotely packaged and digest-verified; the trusted transfer preflight currently fails closed on DNS drift.

Physical errors are post-run diagnostics and are never supplied to either online controller. Observable errors are normalized by the RMS excursion of the corresponding exact trajectory. A completed pair is comparison-admissible when both arms remain within the 15% response guard and absolute safety caps and both their aggregate accuracy burdens and every deadbanded primitive q component agree within a factor of two; the raw errors remain visible below.

Unfinished and transfer-pending evidence — visible for author review

Job	Method	Controller	Drive	τ_{L^2}	Scheduler state	Progress	Available data
9703334.0	AVQDS	state	slowstrong	1.11×10^{-8}	held	—	cgroup-memory hold; no serialized trajectory
9703747.0	AVQDS	state	fastweak	1.11×10^{-8}	held	—	cgroup-memory hold; no serialized trajectory
9703851.0	AVQDS	state	midres	1.11×10^{-8}	held	last: $t = 0.12/5$, $N_\theta = 1960$	now held; no serialized trajectory
9713112.3	AVQDS	composed	strongfast	9×10^{-6}	transfer pending	exit 3	remote run archived; local fetch blocked by trusted DNS guard

These rows expose unfinished or not-yet-ingested attempts rather than hiding them. They are not ledger records, have not been exact-evaluated, and are not accuracy-classified or automatically promoted to the manuscript. Observable errors are intentionally blank until the preserved process-3 archive passes the local transfer and validation transaction.

Resource-rescue completed evidence — cluster 9713112

The two successful screens below were exact-evaluated under the canonical response-normalized policy. The three unsupported-step outcomes are provenance-only safeguards and therefore carry no invented error values.

Job	Method	Controller	Drive	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\max_i \lvert\Delta n_i\rvert}$	$\overline{1 - F}$
9713112.0	AVQDS	composed	fastfast	9.965×10^{-5}	3.683×10^{-5}	4.238×10^{-5}	5.394×10^{-7}
9713112.1	AVQDS	composed	fastweak	—	—	—	—
9713112.2	AVQDS	composed	midres	4.961×10^{-5}	5.681×10^{-5}	4.042×10^{-5}	2.715×10^{-7}
9713112.4	AVQDS	state	strongfast	—	—	—	—
9713112.5	full APM	composed	slowstrong	—	—	—	—

Jobs 9713112.1 and 9713112.5 each exhausted one declared local-subdivision step; job 9713112.4 exhausted 99. They remain visible but are not accuracy evidence. Jobs 9713112.0 and 9713112.2 are qualified t5 screens and may open only the exact comparator-first continuations allowed by the campaign planner.

Current t20 frontier — all 186 verified continuations

Method	Controller	Drive	τ_{L^2}	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\max_i \lvert\Delta n_i\rvert}$	Receipt
AVQDS	parameter	fastfast	1×10^{-6}	0.000818	0.000134	0.000157	9698264
AVQDS	state	fastfast	3×10^{-7}	0.00045	7.12×10^{-5}	0.000114	9698264
AVQDS	parameter	fastweak	3×10^{-6}	0.000424	0.000118	0.000193	9698264
AVQDS	parameter	midres	3×10^{-6}	0.000477	0.000143	8.82×10^{-5}	9698264
AVQDS	parameter	slowstrong	3×10^{-6}	0.000199	0.000116	5.86×10^{-5}	9698264
AVQDS	composed	slowstrong	3×10^{-6}	0.000118	0.000107	4.23×10^{-5}	9698264
AVQDS	parameter	strongfast	1×10^{-6}	0.00027	7.44×10^{-5}	7.49×10^{-5}	9698264
AVQDS	parameter	weakslow	3×10^{-6}	2.66×10^{-5}	2.99×10^{-5}	0.000176	9698264
AVQDS	composed	weakslow	3×10^{-6}	2×10^{-5}	2.66×10^{-5}	0.000173	9698264
AVQDS	state	weakslow	1×10^{-6}	0.00015	0.00014	0.0004	9698264
APM	parameter	weakslow	3×10^{-6}	3.92×10^{-5}	0.000217	0.000119	9698264
APM	composed	weakslow	3×10^{-6}	4.5×10^{-5}	0.000171	0.000196	9698264
AVQDS	parameter	fastfast	3×10^{-6}	0.000749	0.000124	0.000146	9701941
AVQDS	state	fastfast	3×10^{-6}	0.000449	7.14×10^{-5}	0.000115	9701941
AVQDS	parameter	fastweak	1×10^{-6}	0.000569	0.000129	0.000208	9701941
AVQDS	parameter	midres	1×10^{-6}	0.000458	0.000143	0.000103	9701941
AVQDS	parameter	slowstrong	1×10^{-6}	0.000199	0.000116	5.85×10^{-5}	9701941
AVQDS	composed	slowstrong	1×10^{-6}	0.000134	0.000102	4.93×10^{-5}	9701941
AVQDS	parameter	strongfast	3×10^{-6}	0.000291	7.97×10^{-5}	7.41×10^{-5}	9701941
AVQDS	parameter	weakslow	1×10^{-6}	2.47×10^{-5}	2.57×10^{-5}	0.000148	9701941
AVQDS	composed	weakslow	1×10^{-6}	1.85×10^{-5}	2.25×10^{-5}	0.000142	9701941
AVQDS	state	weakslow	3×10^{-6}	0.00015	0.00014	0.0004	9701941
APM	parameter	weakslow	1×10^{-6}	4×10^{-5}	8.84×10^{-5}	0.000156	9701941
APM	composed	weakslow	1×10^{-6}	5.55×10^{-5}	0.000132	0.000144	9701941
AVQDS	state	fastfast	1×10^{-6}	0.00045	7.14×10^{-5}	0.000114	9703334
AVQDS	parameter	fastweak	3×10^{-7}	0.000412	0.000114	0.000167	9703334
AVQDS	parameter	slowstrong	3×10^{-7}	0.000216	0.000116	5.62×10^{-5}	9703334
AVQDS	parameter	weakslow	3×10^{-7}	2.39×10^{-5}	2.49×10^{-5}	0.000153	9703334
AVQDS	state	weakslow	3×10^{-7}	0.00015	0.000143	0.000397	9703334
APM	state	slowstrong	3×10^{-7}	0.000557	0.000206	0.00042	9703334
AVQDS	parameter	fastfast	3×10^{-7}	0.000691	0.000114	0.000149	9703334
AVQDS	parameter	midres	3×10^{-7}	0.000458	0.000143	0.000103	9703334
AVQDS	composed	slowstrong	3×10^{-7}	0.000172	0.000104	6.12×10^{-5}	9703334
AVQDS	parameter	strongfast	3×10^{-7}	0.00027	7.44×10^{-5}	7.49×10^{-5}	9703334
AVQDS	composed	weakslow	3×10^{-7}	1.8×10^{-5}	2.21×10^{-5}	0.000149	9703334
APM	parameter	fastweak	3×10^{-7}	0.00056	0.000191	0.000244	9703334
APM	parameter	weakslow	3×10^{-7}	6.4×10^{-5}	0.00011	0.000142	9703334
APM	composed	weakslow	3×10^{-7}	7.36×10^{-5}	0.000147	0.000244	9703334
AVQDS	state	weakslow	1×10^{-7}	0.00015	0.000143	0.000397	9703851
AVQDS	state	fastfast	1×10^{-7}	0.000496	8.02×10^{-5}	0.000121	9703851
AVQDS	parameter	fastweak	1×10^{-7}	0.000915	0.00018	0.000278	9703851
AVQDS	parameter	slowstrong	1×10^{-7}	0.000211	0.000113	5.45×10^{-5}	9703851
AVQDS	parameter	weakslow	1×10^{-7}	2.67×10^{-5}	2.62×10^{-5}	0.000137	9703851
APM	state	slowstrong	3×10^{-6}	0.000476	0.000264	0.000328	9703851
APM	state	weakslow	1×10^{-7}	0.000146	0.00024	0.000429	9703851
AVQDS	state	weakslow	3.33×10^{-8}	0.000149	0.000143	0.000393	9705193
AVQDS	parameter	fastfast	1×10^{-7}	0.000808	0.000133	0.00016	9705193
AVQDS	state	fastfast	3.33×10^{-8}	0.000473	7.57×10^{-5}	0.000117	9705193
AVQDS	parameter	fastweak	3.33×10^{-8}	0.00073	0.000148	0.000236	9705193
AVQDS	parameter	midres	1×10^{-7}	0.00101	0.000203	0.000144	9705193
AVQDS	parameter	slowstrong	3.33×10^{-8}	0.000356	0.00015	9.17×10^{-5}	9705193
AVQDS	composed	slowstrong	1×10^{-7}	0.00018	0.000106	5.35×10^{-5}	9705193
AVQDS	parameter	strongfast	1×10^{-7}	0.000395	8.8×10^{-5}	6.83×10^{-5}	9705193
AVQDS	parameter	weakslow	3.33×10^{-8}	3.05×10^{-5}	2.85×10^{-5}	0.000127	9705193
AVQDS	composed	weakslow	1×10^{-7}	2×10^{-5}	2.27×10^{-5}	0.000131	9705193
APM	parameter	fastweak	3×10^{-6}	0.000479	0.000199	0.000292	9705193
APM	state	slowstrong	1×10^{-6}	0.000413	0.000346	0.000276	9705193
APM	parameter	weakslow	1×10^{-7}	5.58×10^{-5}	0.000115	0.000135	9705193
APM	composed	weakslow	1×10^{-7}	6.97×10^{-5}	0.000107	0.000285	9705193
APM	state	weakslow	3×10^{-6}	0.000124	0.00022	0.000291	9705193
AVQDS	parameter	fastfast	3.33×10^{-8}	0.00109	0.00018	0.000203	9707629
AVQDS	parameter	fastweak	1.11×10^{-8}	0.000919	0.00018	0.000272	9707629
AVQDS	parameter	midres	3.33×10^{-8}	0.00133	0.00025	0.000173	9707629
AVQDS	parameter	slowstrong	1.11×10^{-8}	0.000506	0.000192	0.000127	9707629
AVQDS	composed	slowstrong	3.33×10^{-8}	0.000201	0.000107	5.52×10^{-5}	9707629
AVQDS	parameter	weakslow	1.11×10^{-8}	3.36×10^{-5}	3.08×10^{-5}	0.000125	9707629
AVQDS	composed	weakslow	3.33×10^{-8}	2.28×10^{-5}	2.38×10^{-5}	0.000117	9707629
AVQDS	state	weakslow	1.11×10^{-8}	0.000149	0.000143	0.000393	9707629
APM	state	slowstrong	1×10^{-7}	0.000599	0.000266	0.000298	9707629
APM	parameter	weakslow	3.33×10^{-8}	4.2×10^{-5}	0.000112	0.000139	9707629
APM	state	weakslow	1×10^{-6}	0.000113	0.000158	0.000298	9707629
AVQDS	state	fastfast	1.11×10^{-8}	0.00066	0.000106	0.000134	9707629
AVQDS	parameter	strongfast	3.33×10^{-8}	0.000109	5.73×10^{-5}	5.47×10^{-5}	9707629
APM	parameter	fastweak	1×10^{-6}	0.000174	0.00017	0.000163	9707629
APM	composed	weakslow	3.33×10^{-8}	5.01×10^{-5}	0.000208	0.000189	9707629
AVQDS	state	weakslow	3.7×10^{-9}	0.000149	0.000143	0.000393	9708462

Interpretation. The displayed errors average all $t = 0 \dots 20$ reporting points. The individual target requires each observable error to be at most 5% of its exact RMS response and mean infidelity to be at most 10^{-4}. A pair may also qualify for a cost comparison inside the 15% response guard when both aggregate burdens and every deadbanded primitive q component agree within a factor of two. Cluster labels identify incrementally ingested evidence; "prior" marks a preceding rung retained as evidence.

Current incremental t5 measurements — preserved submission-era decisions

The row-level decision labels preserve the componentwise 10^{-4} rule under which these jobs were submitted. The canonical response-normalized replay is summarized below; the old labels remain visible as provenance only.

Method	Controller	Drive	τ_{L^2}	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\max_i \lvert\Delta n_i\rvert}$	Decision
APM	parameter	fastfast	3×10^{-7}	5.59×10^{-5}	0.00015	0.00012	extend
APM	state	fastfast	3×10^{-7}	0.000848	0.000218	0.000208	extend
APM	parameter	fastweak	3×10^{-7}	7.01×10^{-5}	9.95×10^{-5}	7.43×10^{-5}	qualified
APM	state	fastweak	3×10^{-7}	0.000245	7.76×10^{-5}	0.000108	extend
APM	parameter	midres	3×10^{-7}	0.000144	0.000185	5.8×10^{-5}	extend
APM	state	midres	3×10^{-7}	0.000143	0.000173	6.29×10^{-5}	extend
APM	parameter	slowstrong	3×10^{-7}	0.000109	0.000106	9.32×10^{-5}	extend
APM	state	slowstrong	3×10^{-7}	8.41×10^{-5}	5.7×10^{-5}	8.66×10^{-5}	qualified
APM	parameter	strongfast	3×10^{-7}	0.000647	8.23×10^{-5}	0.000134	extend
APM	state	weakslow	3×10^{-7}	5.15×10^{-5}	9.29×10^{-5}	0.000106	extend
AVQDS	state	slowstrong	3.33×10^{-8}	0.000129	4.83×10^{-5}	4.02×10^{-5}	extend
AVQDS	state	fastweak	3.33×10^{-8}	0.000158	3.11×10^{-5}	5.17×10^{-5}	extend
APM	parameter	fastfast	1×10^{-7}	0.000359	0.000235	0.000173	extend
APM	state	fastweak	1×10^{-7}	0.000191	9.47×10^{-5}	7.34×10^{-5}	extend
AVQDS	state	midres	3.33×10^{-8}	0.000244	4.64×10^{-5}	3.64×10^{-5}	extend
APM	state	fastfast	1×10^{-7}	0.000462	9.46×10^{-5}	9.63×10^{-5}	extend
APM	parameter	midres	1×10^{-7}	0.000298	0.000104	5.84×10^{-5}	extend
APM	state	midres	1×10^{-7}	0.000357	0.000105	7.26×10^{-5}	extend
APM	parameter	slowstrong	1×10^{-7}	0.000144	8.79×10^{-5}	5.25×10^{-5}	extend
APM	parameter	strongfast	1×10^{-7}	0.000189	9.5×10^{-5}	0.000103	extend
APM	state	weakslow	1×10^{-7}	2.05×10^{-5}	5.46×10^{-5}	8.82×10^{-5}	qualified
APM	parameter	fastfast	3.33×10^{-8}	0.000318	0.000221	0.000144	extend
APM	state	fastweak	3.33×10^{-8}	0.000223	9.99×10^{-5}	0.0001	extend
APM	state	fastfast	3.33×10^{-8}	0.000357	0.000205	0.000137	extend
APM	parameter	midres	3.33×10^{-8}	0.000546	0.000121	7.59×10^{-5}	extend
APM	state	midres	3.33×10^{-8}	0.000512	0.000106	9.2×10^{-5}	extend
APM	parameter	slowstrong	3.33×10^{-8}	0.000109	3.72×10^{-5}	9.37×10^{-5}	extend
APM	parameter	strongfast	3.33×10^{-8}	0.000585	8.13×10^{-5}	0.000135	extend
APM	parameter	fastfast	1.11×10^{-8}	0.000357	0.000106	0.000152	extend
APM	state	fastfast	1.11×10^{-8}	0.000922	0.000198	0.00023	extend
APM	state	fastweak	1.11×10^{-8}	0.000224	0.000177	0.000118	extend
APM	parameter	midres	1.11×10^{-8}	0.000141	6.11×10^{-5}	0.000131	extend
APM	state	midres	1.11×10^{-8}	0.000117	5.64×10^{-5}	0.000121	extend

Current t5 ledger (157 rungs).

Recorded status	Rungs	Scientific meaning
accuracy qualified	142	every completed t5 trajectory meets the 5% exact-response target
requires threshold extension	0	no completed t5 trajectory requires another canonical rung
safeguard tripped	15	provenance only; trajectory not used as accuracy evidence

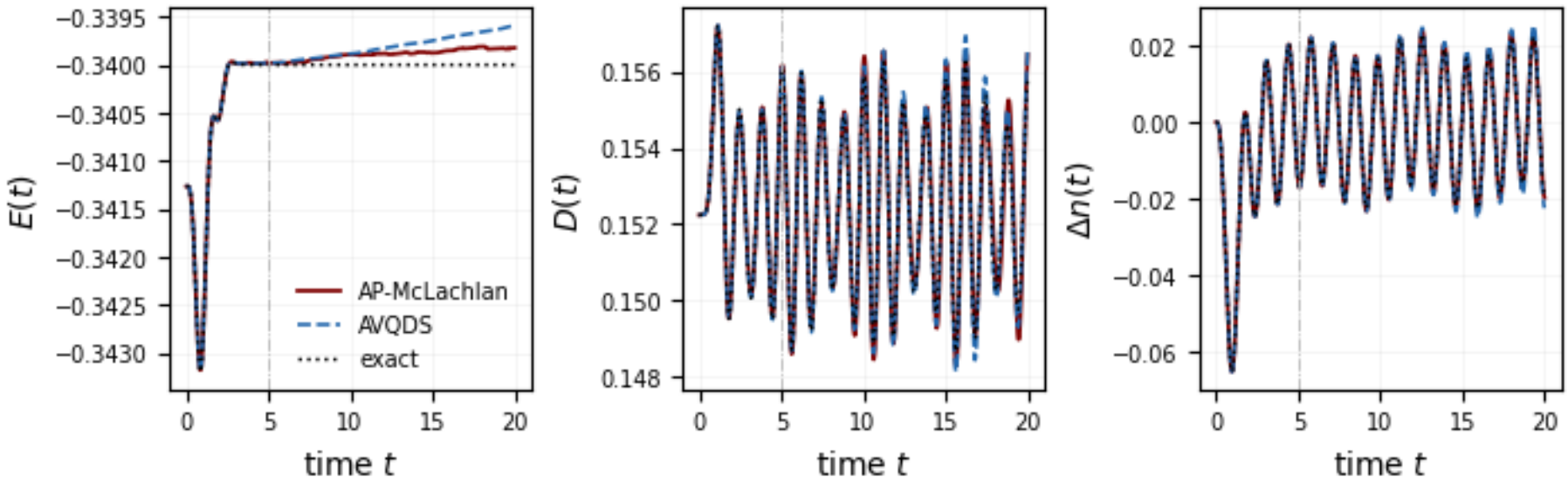
Planned $t = 20$ exact/APM/AVQDS tuple matrix

Every physical-regime, drive, and time-step-controller cell exists before all of its arms finish. One verified exact reference is shared across controller cells for the same physical drive. Every completed arm is shown numerically; only arms without a valid completed trajectory are pending. Qualification uses exact-response-normalized observable errors. Each arm targets at most five percent of the induced RMS response and $\overline{1-F} \le 10^{-4}$. A completed pair may instead be admitted for cost comparison when both arms stay inside the fifteen-percent physical guard and their aggregate accuracy burdens agree within a factor of two, with every deadbanded primitive q component also agreeing within that factor.

Drive	controller	method	$\overline{1-F}$	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\lvert\Delta(\Delta n)\rvert}$	N_{step}	N_θ	$\widehat{\mathbf{C}}_G$
weak–slow	$\delta\theta$	full APM	6.4×10^{-6}	8.6×10^{-5}	2.2×10^{-4}	4.3×10^{-4}	166,918	53	(172, 254, 365)
weak–slow	$\delta\theta$	AVQDS	2.1×10^{-6}	1.5×10^{-4}	1.4×10^{-4}	8.0×10^{-4}	93,154	945	(5214, 5476, 9281)
weak–slow	combined	full APM	6.4×10^{-6}	1.0×10^{-4}	1.9×10^{-4}	3.9×10^{-4}	230,000	53	(178, 248, 395)
weak–slow	combined	AVQDS	1.2×10^{-6}	1.0×10^{-4}	9.7×10^{-5}	5.8×10^{-4}	149,760	926	(5130, 5390, 9048)
weak–slow	state motion	full APM	7.4×10^{-6}	1.3×10^{-4}	1.2×10^{-4}	8.2×10^{-4}	39,408	54	(182, 256, 388)
weak–slow	state motion	AVQDS	1.6×10^{-6}	1.5×10^{-4}	1.4×10^{-4}	7.9×10^{-4}	37,104	370	(1908, 2074, 3642)
fast–weak	$\delta\theta$	full APM	1.1×10^{-5}	5.6×10^{-4}	1.9×10^{-4}	4.9×10^{-4}	698,468	40	(96, 144, 246)
fast–weak	$\delta\theta$	AVQDS	6.0×10^{-7}	4.2×10^{-4}	1.2×10^{-4}	3.9×10^{-4}	796,269	61	(170, 238, 371)
fast–weak	combined	full APM				*not completed*			
fast–weak	combined	AVQDS				*not completed*			
fast–weak	state motion	full APM				*not completed*			
fast–weak	state motion	AVQDS				*not completed*			
mid-resonant	$\delta\theta$	full APM				*not completed*			
mid-resonant	$\delta\theta$	AVQDS	7.2×10^{-7}	4.8×10^{-4}	1.4×10^{-4}	1.8×10^{-4}	947,997	60	(166, 232, 366)
mid-resonant	combined	full APM				*not completed*			
mid-resonant	combined	AVQDS				*not completed*			
mid-resonant	state motion	full APM				*not completed*			
mid-resonant	state motion	AVQDS				*not completed*			
slow–strong	$\delta\theta$	full APM				*not completed*			
slow–strong	$\delta\theta$	AVQDS	6.2×10^{-7}	2.0×10^{-4}	1.2×10^{-4}	1.2×10^{-4}	518,993	62	(186, 248, 396)
slow–strong	combined	full APM				*not completed*			
slow–strong	combined	AVQDS	5.1×10^{-7}	1.2×10^{-4}	1.1×10^{-4}	8.5×10^{-5}	829,568	62	(186, 248, 396)
slow–strong	state motion	full APM	1.4×10^{-5}	5.6×10^{-4}	2.1×10^{-4}	8.4×10^{-4}	250,464	38	(102, 148, 240)
slow–strong	state motion	AVQDS				*not completed*			
strong–fast	$\delta\theta$	full APM				*not completed*			
strong–fast	$\delta\theta$	AVQDS	1.2×10^{-7}	2.7×10^{-4}	7.4×10^{-5}	1.5×10^{-4}	1,000,649	62	(182, 256, 386)
strong–fast	combined	full APM				*not completed*			
strong–fast	combined	AVQDS				*not completed*			
strong–fast	state motion	full APM				*not completed*			
strong–fast	state motion	AVQDS				*not completed*			
fast–fast	$\delta\theta$	full APM				*not completed*			
fast–fast	$\delta\theta$	AVQDS	4.1×10^{-7}	8.2×10^{-4}	1.3×10^{-4}	3.1×10^{-4}	687,685	65	(192, 258, 417)
fast–fast	combined	full APM				*not completed*			
fast–fast	combined	AVQDS				*not completed*			
fast–fast	state motion	full APM				*not completed*			
fast–fast	state motion	AVQDS	1.9×10^{-7}	4.5×10^{-4}	7.1×10^{-5}	2.3×10^{-4}	504,448	100	(316, 408, 672)
insertion McLachlan (IM)					$t = 20$ *matrix pending*				

Promotion preview — weak–slow: $\delta\theta$ controller

Status: verified exact-evaluated comparison evidence. This page is deliberately formatted as a Paper-II result preview. Promotion requires either the individual response target or the declared matched-pair comparison rule.



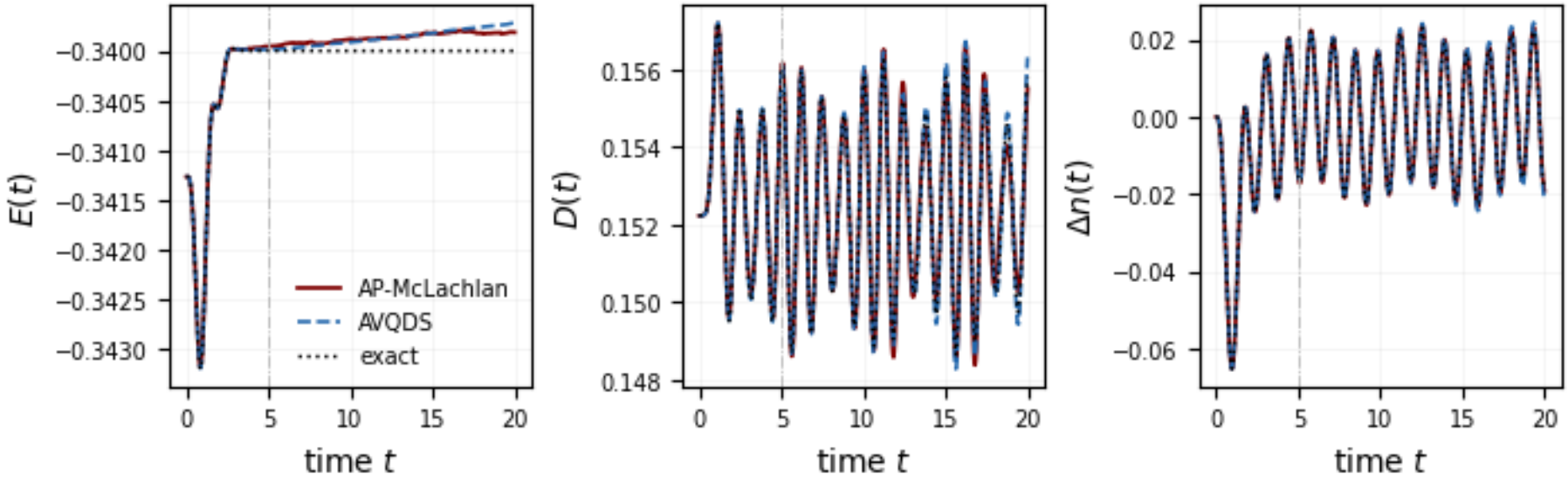
Method	$\overline{1-F}$	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\lvert\Delta(\Delta n)\rvert}$	N_{step}	N_θ	$\widehat{\mathbf{C}}_G = (\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q})$
full APM	6.4×10^{-6}	8.6×10^{-5}	2.2×10^{-4}	4.3×10^{-4}	166,918	53	$(172, 254, 365)$
AVQDS	2.1×10^{-6}	1.5×10^{-4}	1.4×10^{-4}	8.0×10^{-4}	93,154	945	$(5214, 5476, 9281)$
insertion McLachlan (IM)	*t=20 row pending*						

Current qualification gap. full APM: response-normalized pair-matching rule passes; AVQDS: response-normalized pair-matching rule passes.

Exact: dotted black; full APM: solid dark red; AVQDS: dashed blue. Mean infidelity and all three mean observable errors cover the stitched 501-point trajectory on $0 \le t \le 20$. For this fixed-number two-site benchmark, $\overline{\lvert\Delta(\Delta n)\rvert} = 2\,\overline{\max_i \lvert\Delta n_i\rvert}$. The exact reference is post-run and shared by both methods.

Promotion preview — weak–slow: combined controller

Status: verified exact-evaluated comparison evidence. This page is deliberately formatted as a Paper-II result preview. Promotion requires either the individual response target or the declared matched-pair comparison rule.



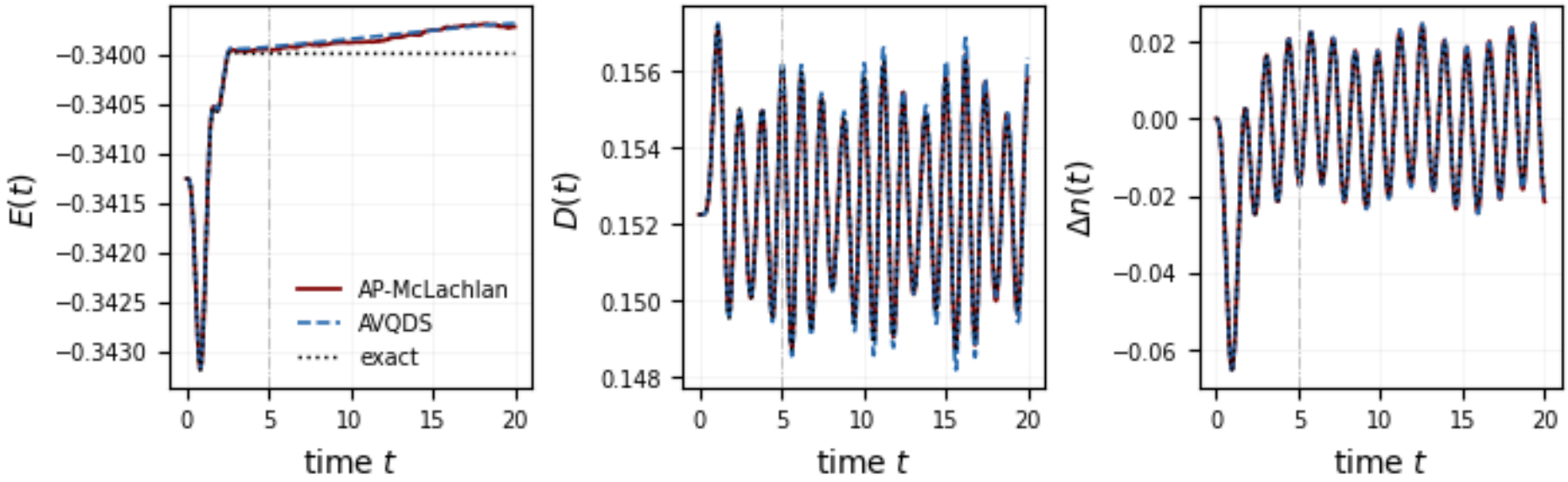
Method	$\overline{1-F}$	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\lvert\Delta(\Delta n)\rvert}$	N_{step}	N_θ	$\widehat{\mathbf{C}}_G = (\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q})$
full APM	6.4×10^{-6}	1.0×10^{-4}	1.9×10^{-4}	3.9×10^{-4}	230,000	53	$(178, 248, 395)$
AVQDS	1.2×10^{-6}	1.0×10^{-4}	9.7×10^{-5}	5.8×10^{-4}	149,760	926	$(5130, 5390, 9048)$
insertion McLachlan (IM)	*t=20 row pending*						

Current qualification gap. full APM: response-normalized pair-matching rule passes; AVQDS: response-normalized pair-matching rule passes.

Exact: dotted black; full APM: solid dark red; AVQDS: dashed blue. Mean infidelity and all three mean observable errors cover the stitched 501-point trajectory on $0 \le t \le 20$. For this fixed-number two-site benchmark, $\overline{\lvert\Delta(\Delta n)\rvert} = 2\,\overline{\max_i \lvert\Delta n_i\rvert}$. The exact reference is post-run and shared by both methods.

Promotion preview — weak–slow: state motion controller

Status: verified exact-evaluated comparison evidence. This page is deliberately formatted as a Paper-II result preview. Promotion requires either the individual response target or the declared matched-pair comparison rule.



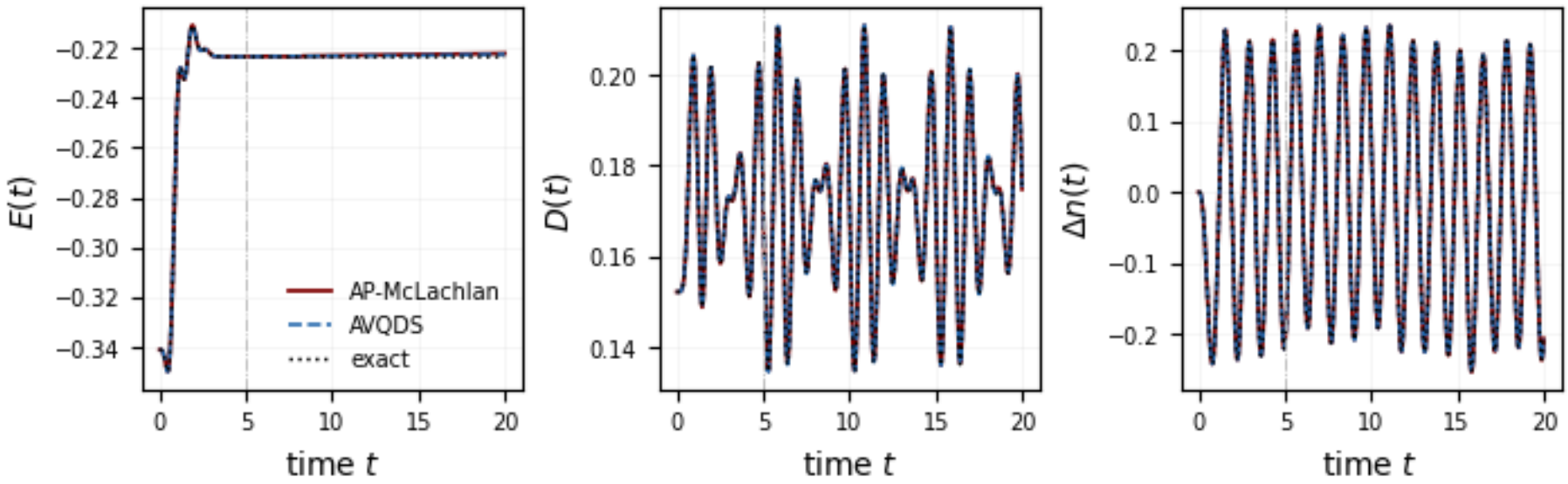
Method	$\overline{1-F}$	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\lvert\Delta(\Delta n)\rvert}$	N_{step}	N_θ	$\widehat{\mathbf{C}}_G = (\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q})$
full APM	7.4×10^{-6}	1.3×10^{-4}	1.2×10^{-4}	8.2×10^{-4}	39,408	54	(182, 256, 388)
AVQDS	1.6×10^{-6}	1.5×10^{-4}	1.4×10^{-4}	7.9×10^{-4}	37,104	370	(1908, 2074, 3642)
insertion McLachlan (IM)	*t=20 row pending*						

Current qualification gap. full APM: response-normalized pair-matching rule passes; AVQDS: response-normalized pair-matching rule passes.

Exact: dotted black; full APM: solid dark red; AVQDS: dashed blue. Mean infidelity and all three mean observable errors cover the stitched 501-point trajectory on $0 \le t \le 20$. For this fixed-number two-site benchmark, $\overline{\lvert\Delta(\Delta n)\rvert} = 2\,\overline{\max_i \lvert\Delta n_i\rvert}$. The exact reference is post-run and shared by both methods.

Promotion preview — fast–weak: $\delta\theta$ controller

Status: verified exact-evaluated comparison evidence. This page is deliberately formatted as a Paper-II result preview. Promotion requires either the individual response target or the declared matched-pair comparison rule.



Method	$\overline{1-F}$	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\lvert\Delta(\Delta n)\rvert}$	N_{step}	N_θ	$\widehat{\mathbf{C}}_G = (\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q})$
full APM	1.1×10^{-5}	5.6×10^{-4}	1.9×10^{-4}	4.9×10^{-4}	698,468	40	$(96, 144, 246)$
AVQDS	6.0×10^{-7}	4.2×10^{-4}	1.2×10^{-4}	3.9×10^{-4}	796,269	61	$(170, 238, 371)$
insertion McLachlan (IM)	*t=20 row pending*						

Current qualification gap. full APM: individual response target passes; AVQDS: individual response target passes.

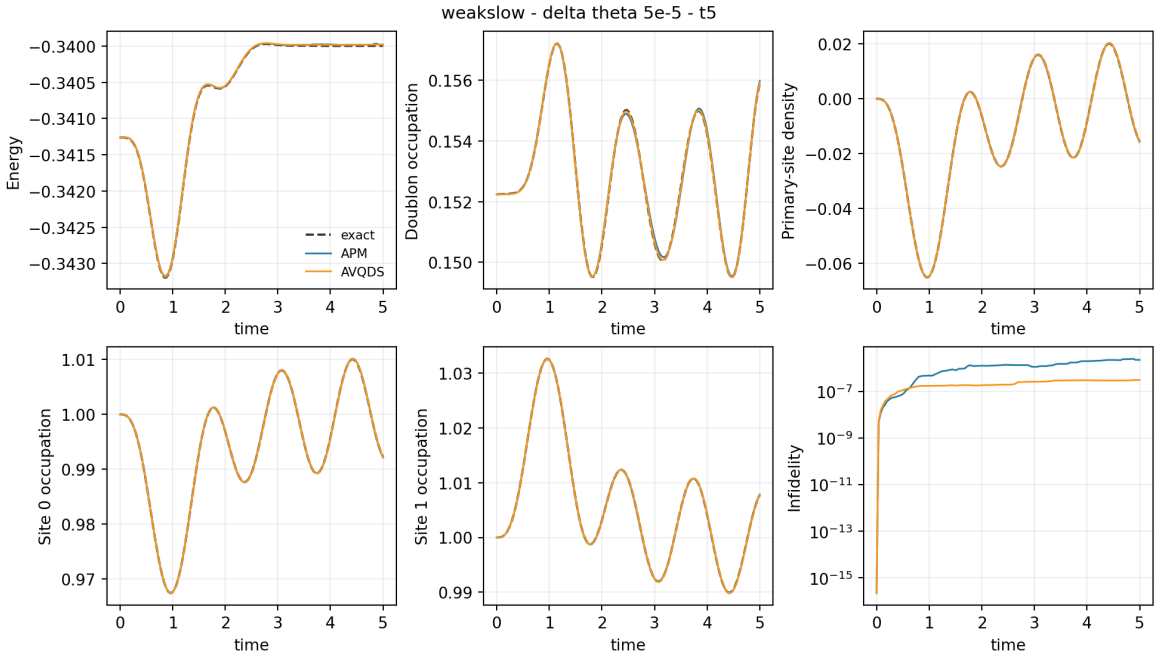
Exact: dotted black; full APM: solid dark red; AVQDS: dashed blue. Mean infidelity and all three mean observable errors cover the stitched 501-point trajectory on $0 \le t \le 20$. For this fixed-number two-site benchmark, $\overline{\lvert\Delta(\Delta n)\rvert} = 2\,\overline{\max_i \lvert\Delta n_i\rvert}$. The exact reference is post-run and shared by both methods.

Archived pre-change t5 screening pairs

Source ledger: `output/paper_ii_staged/fetched/cluster_9698201.final22/ledger_corrected_final22.json` **Qualified pairs:** 2 **Ledger rows:** 47 **Audit:** PASS **Current role:** historical screening evidence, replayed under the canonical response-normalized policy

Costs are reported without an automatic method judgment.

weakslow - delta theta 5e-5 - t5



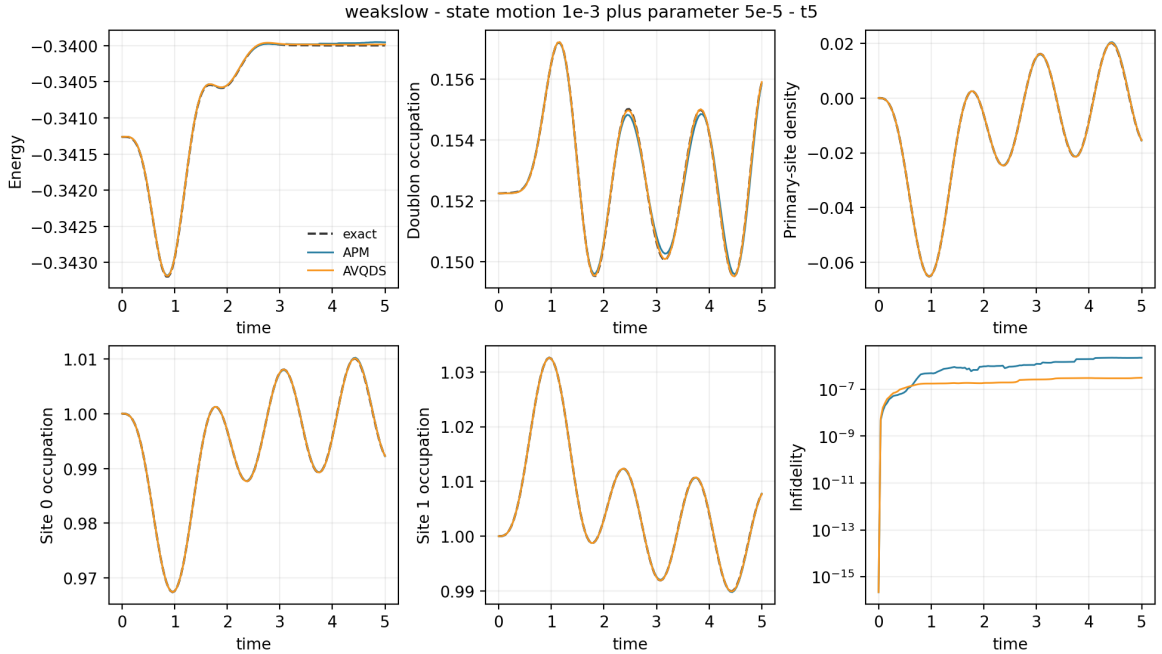
Physical errors against the exact diagnostic trajectory.

Method	τ_{L^2}	$\langle\lvert\Delta E\rvert\rangle$	$\langle\lvert\Delta n_d\rvert\rangle$	$\langle\max_i \lvert\Delta n_i\rvert\rangle$	mean infidelity
APM	3.000000e-06	1.164870e-05	5.802233e-05	4.634581e-05	1.243413e-06
AVQDS	3.000000e-06	1.075525e-05	1.738386e-05	3.261893e-05	2.192390e-07

Terminal resources and integration work.

Method	$\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q}, N_\theta$	$\Delta\widehat{C}_{2q}, \Delta\widehat{C}_D, \Delta\widehat{C}_{1q}, \Delta N_\theta$	steps acc./att.	RHS	ins./del./exch.
APM	(124, 178, 292, 42)	(+60, +70, +144, +14)	132151/132151	132151	48/33/3
AVQDS	(156, 218, 340, 56)	(+92, +110, +192, +28)	74937/74937	74937	28/0/0

weakslow - state motion 1e-3 plus parameter 5e-5 - t5



Physical errors against the exact diagnostic trajectory.

Method	τ_{L^2}	$\langle \lvert \Delta E \rvert \rangle$	$\langle \lvert \Delta n_d \rvert \rangle$	$\langle \max_i \lvert \Delta n_i \rvert \rangle$	mean infidelity
APM	3.000000e-06	1.510702e-05	9.128360e-05	6.692100e-05	1.123767e-06
AVQDS	3.000000e-06	9.511439e-06	1.699246e-05	3.241321e-05	2.172700e-07

Terminal resources and integration work.

Method	$\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q}, N_\theta$	$\Delta\widehat{C}_{2q}, \Delta\widehat{C}_D, \Delta\widehat{C}_{1q}, \Delta N_\theta$	steps acc./att.	RHS	ins./del./exch.
APM	(112, 156, 248, 38)	(+48, +48, +100, +10)	182192/205501	205501	44/33/3
AVQDS	(156, 218, 340, 56)	(+92, +110, +192, +28)	124032/139261	139261	28/0/0

Gate-pending completed diagnostics — cluster 9712867

All eight trajectories completed and were evaluated against the shared exact-reference store. Their configured solve-repair ridge ladder, however, differs from the declared numerical policy, so these measurements are retained as numerical diagnostics and are not canonical ledger records or manuscript promotions. A failed configuration guard does not erase completed data; the affected actions must be reissued under the locked policy.

Job	Method	Controller	Drive	$\overline{\lvert\Delta E\rvert}$	$\overline{\lvert\Delta D\rvert}$	$\overline{\max_i \lvert\Delta n_i\rvert}$	$\overline{1-F}$
9712867.0	AVQDS	state	fastweak	5.253×10^{-4}	1.515×10^{-4}	2.302×10^{-4}	9.577×10^{-7}
9712867.1	AVQDS	state	midres	5.833×10^{-4}	1.693×10^{-4}	1.372×10^{-4}	8.889×10^{-7}
9712867.2	AVQDS	state	slowstrong	4.138×10^{-4}	1.807×10^{-4}	1.296×10^{-4}	8.626×10^{-7}
9712867.3	full APM	parameter	fastfast	3.777×10^{-4}	2.770×10^{-4}	3.585×10^{-4}	9.231×10^{-6}
9712867.4	full APM	state	fastfast	8.104×10^{-4}	2.245×10^{-4}	4.656×10^{-4}	1.190×10^{-5}
9712867.5	full APM	parameter	midres	5.255×10^{-4}	3.197×10^{-4}	1.653×10^{-4}	4.949×10^{-6}
9712867.6	full APM	parameter	slowstrong	6.925×10^{-4}	5.219×10^{-4}	3.858×10^{-4}	4.145×10^{-5}
9712867.7	full APM	parameter	strongfast	3.228×10^{-4}	2.387×10^{-4}	4.851×10^{-4}	5.198×10^{-6}

Disposition: **gate pending — canonical reissue required**. The measured state-motion intervals used the declared base ridge and no uncured local subdivisions were recorded, but immutable configuration equality is required before ledger admission. This page is author-review evidence only and does not alter the 18-tuple/36-slot manuscript matrix.

SIAM screening benchmark results

Role: diagnostic screening evidence - not manuscript promotion | grid: 101 points, t=0..1 cell 6030dc0e971a

Method	State	m_d	D_d	C_1	C_2	C_3	C_4	mean 1-F	g_E	steps	N_theta	(C2q,CD,C1q)
Exact	**reference**	0	0	0	0	0	0	0	0	--	--	--
AVQDS	**candidate**	2.001e-03	3.574e-03	3.526e-03	9.980e-03	7.938e-02	2.274e-02	2.602e-05	8.603e-05	25,600	24	(370,704,687)
Full APM	**candidate**	1.809e-04	9.211e-04	4.599e-03	6.179e-03	3.190e-01	3.524e-04	2.237e-06	5.749e-05	25,600	30	(912,1432,1254)

Pair verdict: accuracy-confounded.

Errors are declared-grid (t=0 to 1) response-normalized MAEs; candidates retain numerical values.

Canonical resources are logical N_theta and the deterministic graph proxy; no Qiskit compilation is used.

SIAM $N_b = 4$ observable screening snapshot

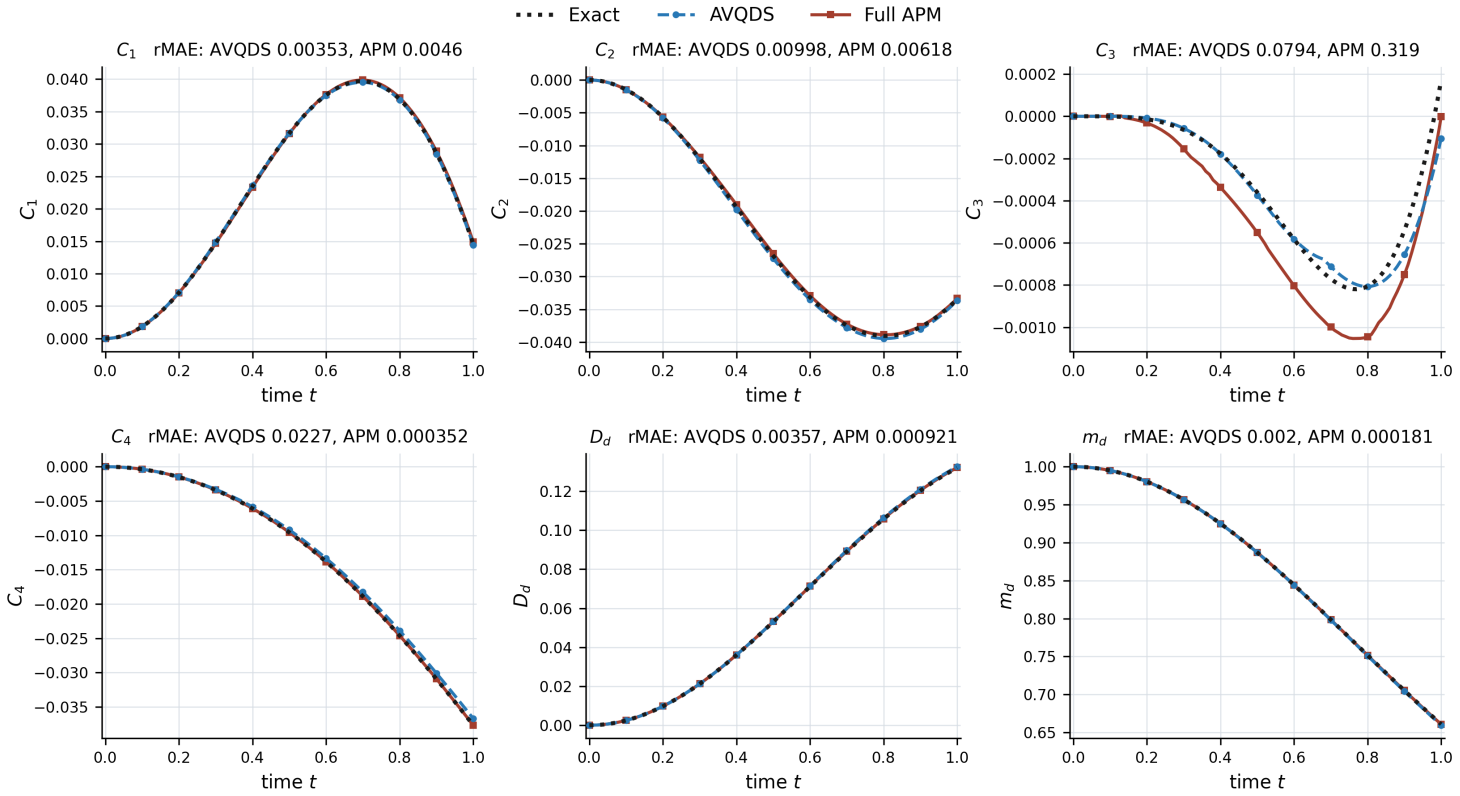
hybridization quench, $t \in [0, 1]$

DIAGNOSTIC ONLY — CURRENT PAIR REJECTED AS ACCURACY-CONFOUNDED; NO COST COMPARISON IS ADMISSIBLE

Source aggregate	Row count: 2 locked exact-scored trajectory JSONs: `sources/siam_avqds_cluster9714484_score.json` (SHA-256 `8c6125f7...35614614`) and `sources/siam_apm_cluster9715147_score.json` (SHA-256 `25414cc6...19a6f2e7`).
Scheduler identities	AVQDS `9714484.0`, cut 10^{-3}; Full APM `9715147.0`, cut 10^{-5}, exchange policy. Each score contains 101 shared time samples on $t = 0, 0.01, \ldots, 1$. Evidence revision `d5590449`; run code `36c40d52`.
Identity audit	**PASS_DIAGNOSTIC_COMPARISON_IDENTITY**: identical grid, seed, Hamiltonian/physics fingerprint, comparison fingerprint, observable contract, exact observable arrays, and exact-energy arrays.
Observable definitions	$m_d = \langle n_{d\uparrow} - n_{d\downarrow} \rangle$, $D_d = \langle n_{d\uparrow} n_{d\downarrow} \rangle$, $C_j = \langle \mathbf{S}_d \cdot \mathbf{S}_j \rangle$ including transverse exchange, and Sigma-v2 $\Sigma_r = \langle S_d^2 \rangle + \sum_{j=1}^{r} C_j$, with $\langle S_d^2 \rangle = \frac{3}{4}(N_d - 2D_d)$.
Interpretation	Exact evolution is reporting-only and was not used by either controller. Neither trajectory passes the individual 5% primitive-response target. Full APM fails the 15% broad guard in the tiny-response C_3 channel, so the pair is accuracy-confounded. Qiskit compilation was not performed.

SIAM primitive observable trajectories

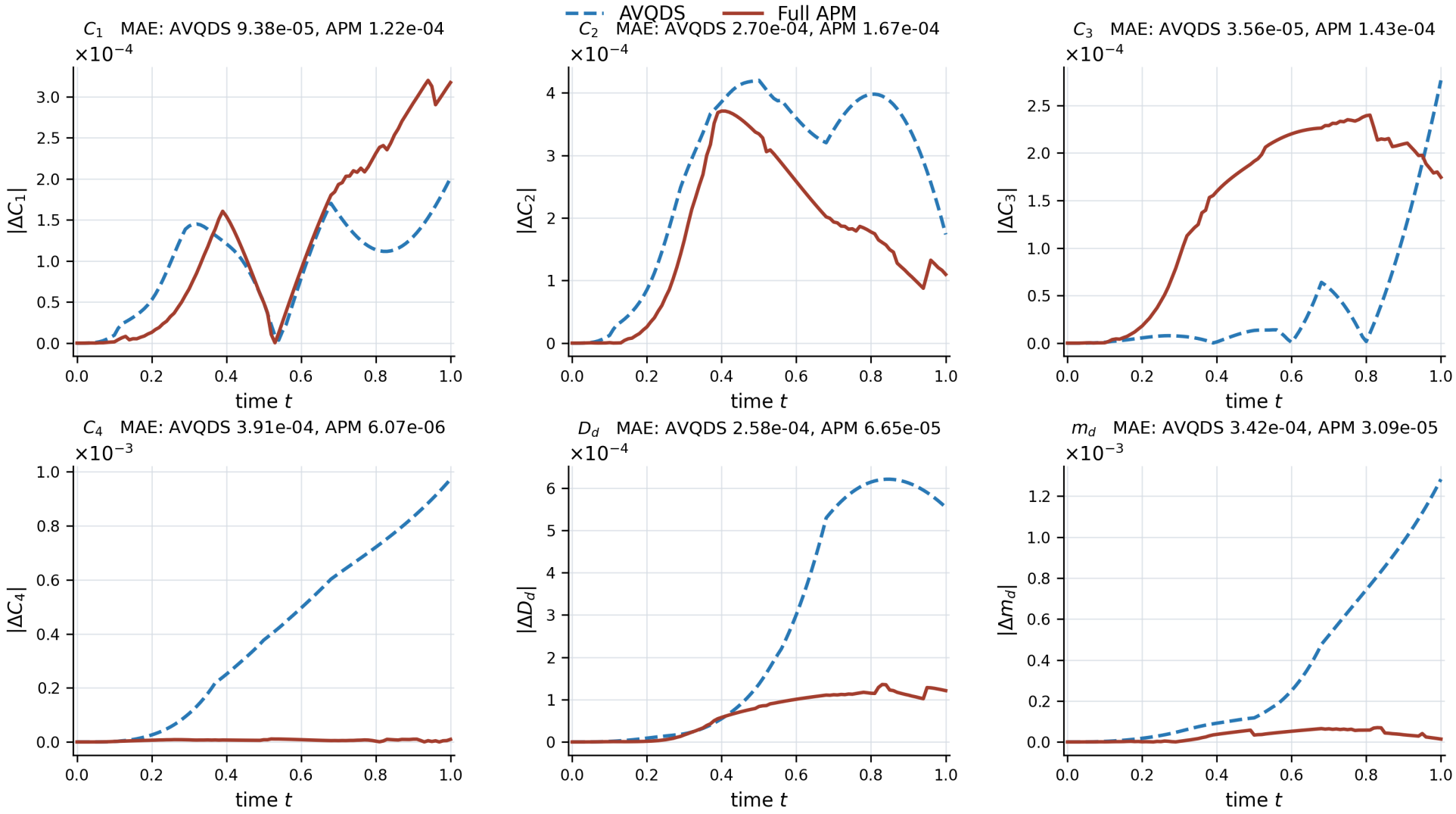
$N_b = 4$, 101 shared points on $t \in [0, 1]$ | AVQDS 9714484 (cut 10^{-3}) | Full APM 9715147 (cut 10^{-5}; exchange policy)



Diagnostic exact-reference comparison only | pair verdict: rejected, accuracy-confounded | exact data were not used by either controller

SIAM primitive observable absolute errors

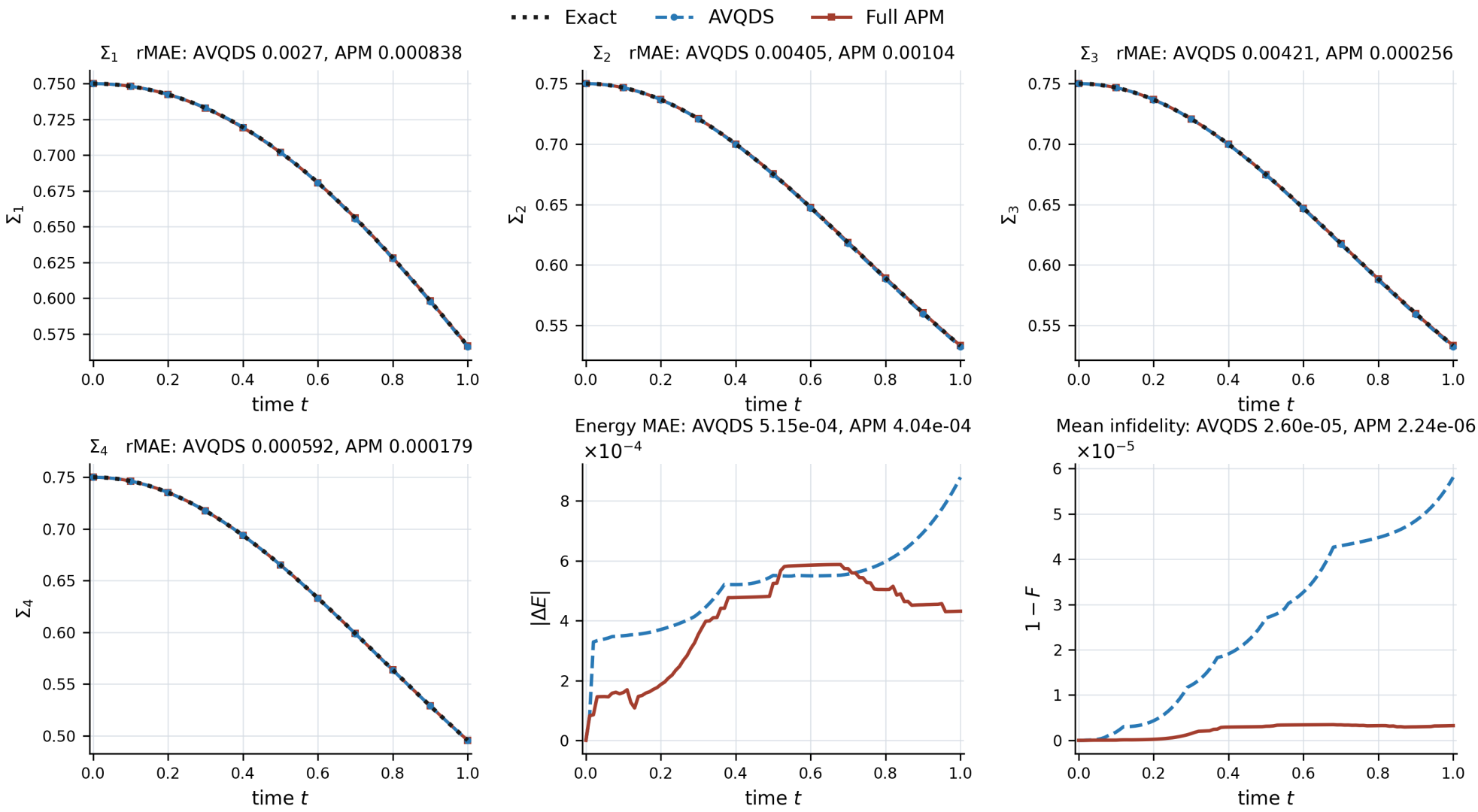
Physical error against the shared exact diagnostic trajectory; each panel reports the time-mean absolute error



Diagnostic exact-reference comparison only | pair verdict: rejected, accuracy-confounded | exact data were not used by either controller

SIAM derived observables and accuracy diagnostics

Derived screening observables Σ_1--Σ_4, physical energy error, and state infidelity



Diagnostic exact-reference comparison only | pair verdict: rejected, accuracy-confounded | exact data were not used by either controller